Computer Science & IT

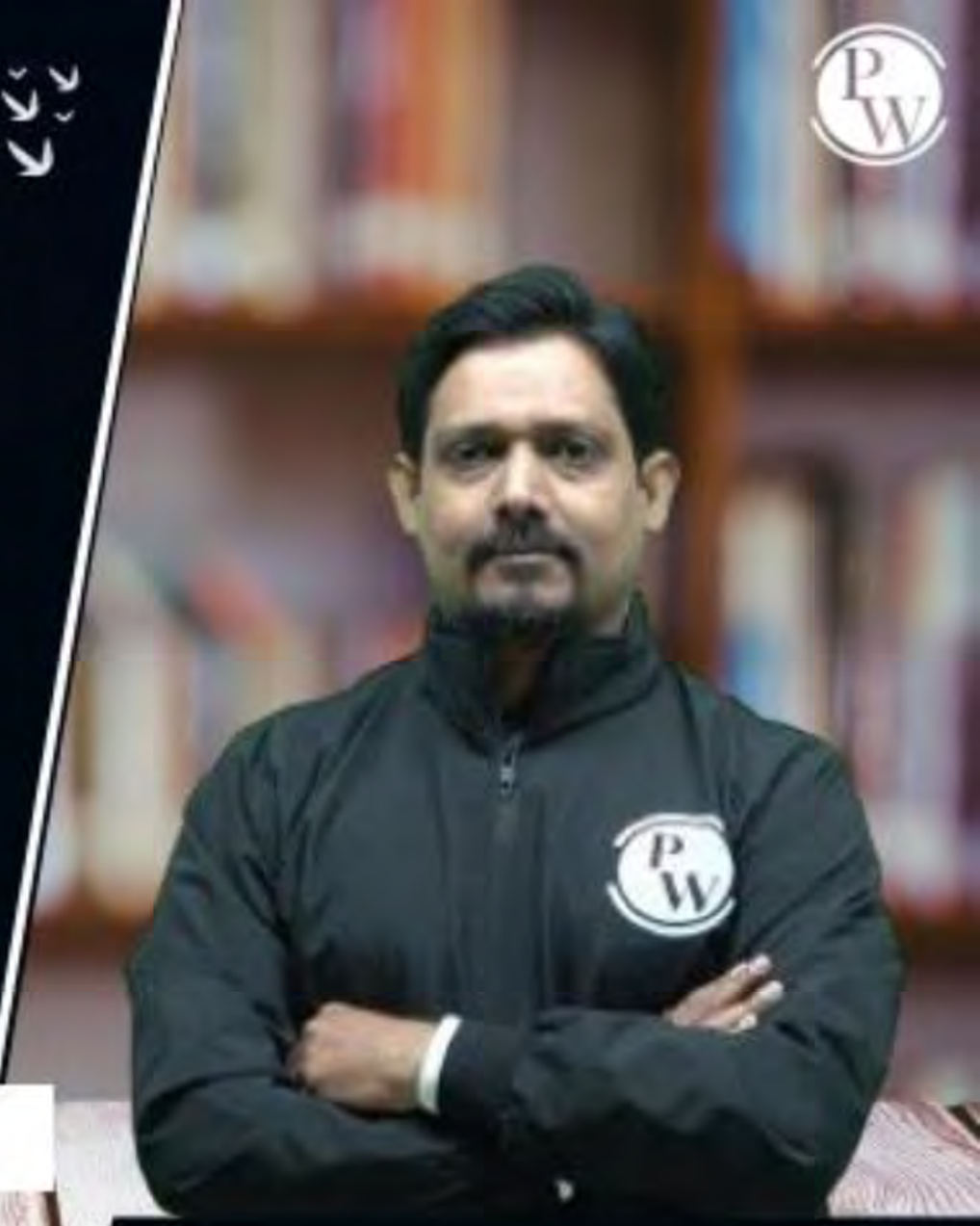
MTA



Data Structure & programming

Linked List

Lecture No. 04



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Recap of Previous Lecture



- Topic — Single linked insertion
- Topic — Deletion
- Topic — problem Solving
- Topic
- Topic

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Topics to be Covered



- Topic — Doubly linked list
- Topic — Add begin
- Topic — Addend
- Topic — Delete in Double
- Topic — Insert in Doubly linked list.

Slide

Topic : Doubly Linked List





NULL	X	NULL

```
typedef struct dnode {
        int data;
        struct dnode * next;
        struct dnode * prev;
} dlnode;
```

```
dnode * getnode(int x) {
  dnode *temp;
  temp = malloc(sizeof(dnode));
  temp->data x;
   temp->next = NULL;
   temp->prev = NULL;
  return temp;
}
```

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Topic : Doubly Linked List



Head pointer HP



pointer

```
dnode* build123() {
    dnode * temp1, *temp2, *temp3;
    temp1 = getnode(1);
    temp2 = getnode(2);
    temp3 = getnode(3)
    temp1 → next = temp2;
    temp2 → next = temp3;
    temp2 → prev = temp1
    temp3 → prev = temp2
    return temp1;
}
```



Slide

Topic : Doubly Linked List






```
int main() {
dnode* HP;

HP = Build123()
.
.
.
}
```

```
dnode* Addbegin( dnode *S, int x){
    dnode* p = getnode(x)
    if (S==NULL)
        return p;
    p->next = S;
    S->prev = p;
    return p;
}
```

Slide

Topic : Doubly Linked List






```
int main() {
dnode* HP;

HP = Build123();
HP = Addbegin(HP,4);

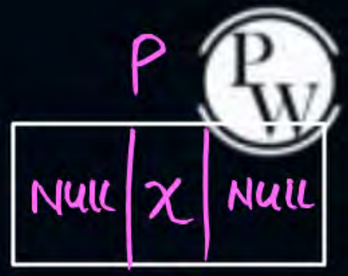
}
```

```
dnode* Addbegin( dnode*s, int x){
    dnode* p = getnode(x);
    p→next = s;
    if(s) s→prev = p;
    return p;
}
```

Slide

Topic : Doubly Linked List





Head pointer




```
dnode * Addend (dnode*s,  int x) {
    dnode * p = getnode(x);
    dnode * q = s;
    if (s == NULL)
        return p;
    while( q->next != NULL)
        q = q->next;
    q->next = p;
    p->prev = q;
    return s;
}
```

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Topic : Doubly Linked List



HP = NULL

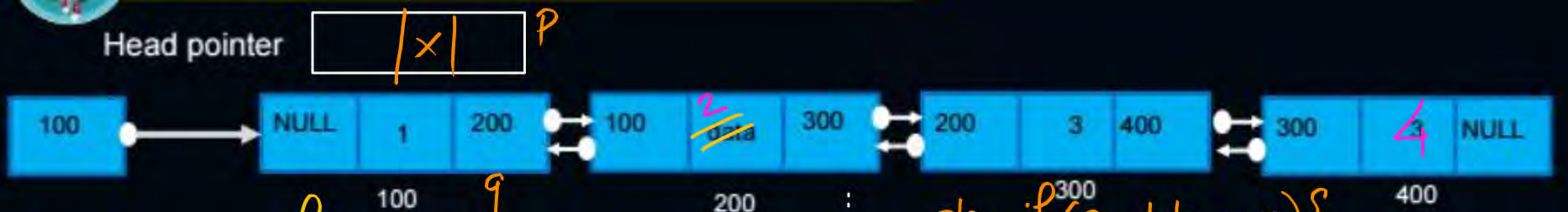
for (int i = 1; i <= 5; i++) ← 5 O(1)

HP = Addend(HP, i);

1 → 2 → 3 → 4 → 5

1 → 2 → 3 → 4 → 5

Slide

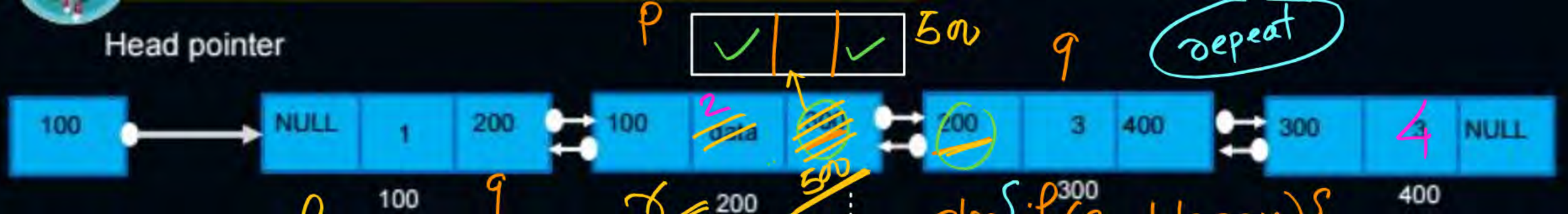


Insert x before Node y

q

```
dnode * Addxbefory (dnode*s, intx, inty)
{
    dnode *q=s, *r=NULL;
    dnode *p = getnode(x);
    if (s==NULL){
        free(p);
        return s;
    }
    else if (q->data==y){
        p->next = q;
        q->prev = p;
        return p;
    }
    while ( q->next!=NULL && q->data!=y)
        q = q->next;
```

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Insert x before Node y

```
dnode* Addxbefory(dnode*s, intx, inty){
  dnode*q=s, *r=NULL;
  dnode*p = getnode(x);
  if(s==NULL){
      free(p);
      return s;
  }
  else{ if(q->data==y){
      p->next=q;
      q->prev=p;
      return p; }
  while(q->next!=NULL && q->data!=y)
      q=q->next;
  if(q->data==y){
      r=q->prev;   r->next=p;
      p->prev=r;   p->next=q;
      q->prev=p; }
  else { printf("y does not exist");
      free(p);
  }
  return s; } }
```

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* if list is empty, y does not exists Hence return

2. first Node contains y



Topic Question

If the list contain the value 1→2→3 then what is the Output following function.

```
void foo (Node *ptr){
if (ptr){
        printf ("%d", ptr ->data);
        fun (ptr -> link);
        printf ("%d", ptr ->data);
        fun (ptr->link);
        return;
}
The number of values printed by the program
is _____
```

No. of value for list having n element = T(n)

1→2

$T(1) = 2$

$T(2) = 2T(1) + 2 = 2 \times 2 + 2 = 6$

$T(3) = 2T(2) + 2$

$= 2 \times 6 + 2 = 14$

Slide

Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node{
        int value;
        struct Node *next;};
```

Consider the following ANSI C program:

```
int main () {
 Struct Node *boxE, *head, *boxN; int index = 0;
 boxE = head = (struct Node*) malloc (sizeof (struct Node));
 head->value = index;
 for (index = 1; index <=3; index++) {
   box N = (struct Node *) malloc (sizeof (struct Node));
   BoxE->next = boxN;
   boxN->value = index;
   boxE= boxN; }

 for (index = 0; index <=3; index++) {
      printf( Value at index %d is %d\n , index, head->value);
      head = head->next;
      printf ( Value at index %d is %d\n , index + 1, head- >value
 }}
```

Which one of the statements below is correct about the program?

(A) It dereferences an uninitialized pointer that may result in a run-time error

(B) It has a missing return which will reported as an error by the compiler

(C) Upon execution, the program creates a linked-list of five nodes

(D) Upon execution, the program goes into an infinite loop

The following C function takes a simply-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

```
typedef struct node {
        int value;
        struct node *next;
}Node;
```

Let LIST be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
int data;
struct list *next;
} LIST;
```

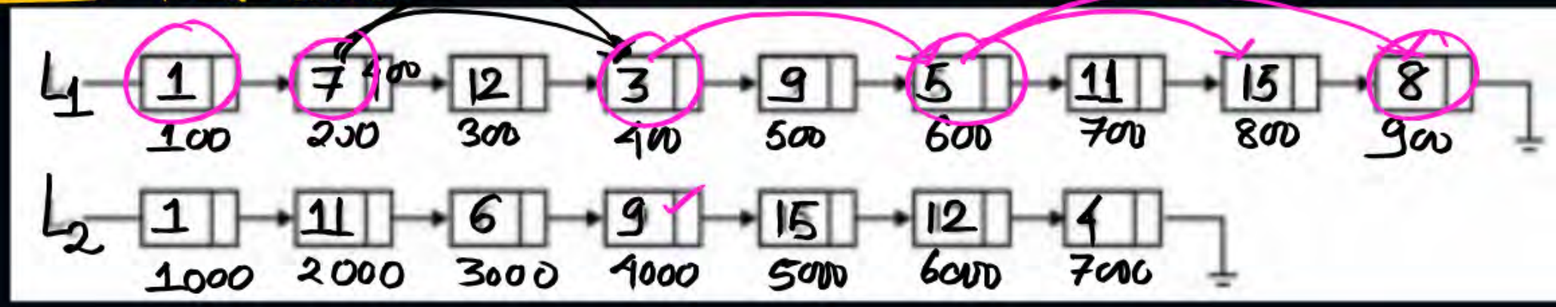
Suppose a program has created 2 linked-lists L1 & L2, whose contents are given in the figure below (code for creating L1 & L2 is not provided here). L1 contains 9 nodes and L2 contains 7 nodes.

Consider the following C-prog segment that modifies the list L1. The number of nodes that will b
there in L1 after execution of the code sequence is _____.

```
int find (int query, LIST *list) {
        while (list != NULL) {
                if (list → data == query) return 1;
                list = list → next;
}
return 0;}
int main (){
        List ptr1 = L1; ptr2 = L2;
        while (ptr1 → next! = NULL) {
                query = ptr1 → next → data;
                if ((find(query, L2))
                                ptr1 → next = ptr1 → next → next;
                else
                                ptr1 = ptr1 → next;
        }
return 0;
}
```

Deadlock

ptr1 = 100 200 400 600
ptr2 = 1000



query : 7 , query : 12 query : 3 , query : 9, query : 5

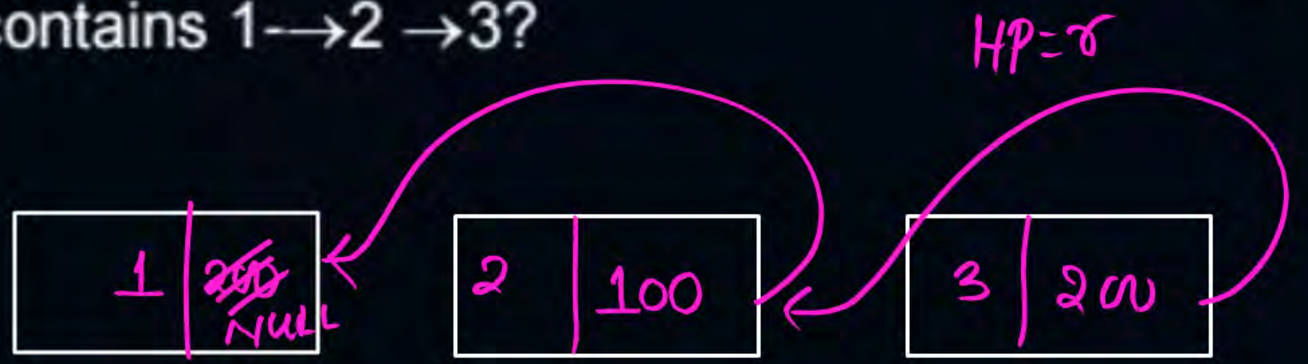
query : 11 query : 15

Question

What is output of the program if the list contains 1⟶2 ⟶3?

```
HP is global variable
void Dosomething(){
        node *q, *r, *s;
        q = HP;
        r = NULL;
              while( q!=NULL){
                      s = r;
                      r = q;
                      q = q→ link;
                      r→ link =s;
               }
               HP = r;
        }
```

3→2→1



q

r=NULL

S=NULL

r=q

S

r

S

q

r

q



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Double linked list

Add begin Addend

Insert x before y

problem

Slide

THANK - YOU